

Alexandra G. Roberts, PhD

✉ alexandrageroberts@gmail.com

📧 agr78

✂ @alexandragrob

🌐 alexandragroberts



Employment

- 2026 **Postdoctoral Research Fellow** Athinoula A. Martinos Center for Biomedical Imaging
Massachusetts General Hospital & Harvard Medical School, Boston MA
- 2017 - 2020 **Research Engineer** Benz Research & Development, Sarasota, FL
- 2014 - 2015 **Corps Member** Teach For America, New York, NY

Education

- 2020 - 2026 **Doctor of Philosophy, Cornell University** Electrical & Computer Engineering
Signal Modeling in Quantitative Susceptibility Mapping: Reconstruction and Applications
- 2016 - 2020 **Master of Engineering, University of Florida** Chemical Engineering
3D Reconstruction Algorithms for X-ray Tomography of Printed Circuit Board Samples
- 2010 - 2014 **Bachelor of Science, West Virginia Wesleyan College** Chemistry
Summa Cum Laude

Personal

Licensed amateur radio operator (KC0AGR), marathon runner, and guitarist.

Research Publications

Selected Articles

- 1 A. G. Roberts et al., "Spectral Vision Transformer for Efficient Tokenization with Limited Data", *Preprint*, May 2026. arXiv: 2605.12026 [cs.CV]. 🔗 URL: <https://arxiv.org/abs/2605.12026>
- 2 A. G. Roberts et al., "Synthetic Generation and Latent Projection Denoising of Rim Lesions in Multiple Sclerosis", *Synthetic Data for Computer Vision Workshop @ CVPR 2025*, Jun. 2025. 🔗 URL: <https://openreview.net/forum?id=wFkiqB5spT>
- 3 A. G. Roberts et al., "Technical Feasibility of Quantitative Susceptibility Mapping Radiomics for Predicting Deep Brain Stimulation Outcomes in Parkinson's Disease", *Neurosurgery*, Feb. 2025. 🔗 DOI: 10.1227/neu.0000000000003721
- 4 A. G. Roberts et al., "Radiomic Prediction of Parkinson's Disease Deep Brain Stimulation Surgery Outcomes using Quantitative Susceptibility Mapping and Label Noise Compensation", *Brain Stimul.*, vol. 18, no. 4, pp. 1286-1288, Oct. 2024. 🔗 DOI: 10.1016/j.brs.2025.05.062
- 5 A. G. Roberts et al., "Maximum Spherical Mean Value Filtering for Whole-brain QSM", *Magnetic Resonance in Medicine*, vol. 91, no. 4, pp. 1586-1597, Jan. 2024, ISSN: 1522-2594. 🔗 DOI: 10.1002/mrm.29963

Conference Proceedings

- 1 A. G. Roberts et al., "Joint Prediction of Motor and Non-motor Deep Brain Stimulation Outcomes using Quantitative Susceptibility Mapping", in *Movement Disorders Congress*, Oct. 2025.
- 2 A. G. Roberts, J. Zhang, P. Spincemaille, T. Nguyen, and Y. Wang, "Radiomic prediction of Parkinson's disease deep brain stimulation surgery motor and nonmotor outcomes using quantitative susceptibility mapping", in *Magnetic Resonance Phase, Susceptibility, and Electrical Properties Mapping*, 2024.
- 3 A. G. Roberts, A. Dimov, T. Nguyen, P. Spincemaille, and Y. Wang, "Inhomogeneity-informed field-fitting for quantitative susceptibility mapping (if-QSM)", in *Proceedings of the 33rd Annual Meeting of ISMRM, Singapore*, 2024.
- 4 A. G. Roberts et al., "QRadAR: A toolbox for quantitative magnetic resonance radiomics analysis and reliability", in *Proceedings of the 33rd Annual Meeting of ISMRM, Singapore*, 2024.
- 5 A. G. Roberts et al., "Radiomics for deep brain stimulation outcome prediction using quantitative susceptibility mapping (RadDBS-QSM)", in *Proceedings of the 33rd Annual Meeting of ISMRM, Singapore*, 2024.
- 6 A. G. Roberts et al., "Whole brain source separation for neurodegeneration", in *Proceedings of the 33rd Annual Meeting of ISMRM, Singapore*, 2024.
- 7 A. Roberts, P. Spincemaille, T. Nguyen, and Y. Wang, "Whole brain spherical mean value filtering for shadow reduction in quantitative susceptibility mapping", in *Proceedings of the 32nd Annual Meeting of ISMRM, Toronto, Canada*, 2023.
- 8 A. G. Roberts, I. Kovanlikaya, B. Kopell, P. Spincemaille, T. Nguyen, and Y. Wang, "Improved visualization of the medial medullary lamina via phase priors in quantitative susceptibility mapping", in *Proceedings of the 32nd Annual Meeting of ISMRM, Toronto, Canada*, 2023.
- 9 A. G. Roberts, J. Zhang, P. Spincemaille, T. Nguyen, and Y. Wang, "Choice of architecture in 3d super-resolution of quantitative susceptibility maps", in *Proceedings of the 32nd Annual Meeting of ISMRM, Toronto, Canada*, 2023.
- 10 A. Roberts, P. Spincemaille, I. Kovanlikaya, J. A. Tsiouris, T. Nguyen, and Y. Wang, "SWISer: Multi-field susceptibility-weighted images super-resolution", in *Proceedings of the 31st Annual Meeting of ISMRM, London, England*, 2022.
- 11 A. G. Roberts, J. Zhang, P. Spincemaille, T. Nguyen, and Y. Wang, "SuperQ: 3d super-resolution of quantitative susceptibility maps", in *Proceedings of Magnetic Resonance Phase, Susceptibility, and Electrical Properties Mapping*, 2022.
- 12 A. G. Roberts, P. Spincemaille, T. Nguyen, I. Kovanlikaya, and Y. Wang, "MEDI-FM: Field map error guided regularization for shadow reduction in quantitative susceptibility mapping", in *Proceedings of the 31st Annual Meeting of ISMRM, London, England*, 2022.
- 13 A. Roberts, P. Spincemaille, T. D. Nguyen, and Y. Wang, "MEDI-d: Downsampled morphological priors for shadow reduction in quantitative susceptibility mapping", in *Proceedings of the 30th Annual Meeting of ISMRM*, 2021.
- 14 A. Roberts, J. True, N. T. Jessurun, and D. N. Asadizanjani, "An overview of 3d X-Ray reconstruction algorithms for PCB inspection", ASM International, Dec. 2020.  DOI: 10.31399/asm.cp.istfa2020p0188
- 15 A. Roberts, D. Russell, and B. Anthony, "Systems genomics in alcohol exposed dorsal root ganglia neuronal stem cells of adult rats", in *Proceedings of the West Virginia Academy of Sciences*, 2014.